

# | Equation of Motion Time Discretization

| Newton's 2nd Law as a  
first-order system

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{v} \\ \mathbf{M}\dot{\mathbf{v}} = \mathbf{f}(\mathbf{x}, \mathbf{v}) \end{cases}$$

BDF1

$$\dot{\mathbf{y}}_t \approx \frac{\mathbf{y}_{t+1} - \mathbf{y}_t}{h}$$

| Explicit Euler

$$\begin{cases} \mathbf{x}^+ = \mathbf{x}^- + h \mathbf{v}^- \\ \mathbf{v}^+ = \mathbf{v}^- + h \mathbf{M}^{-1} \mathbf{f}(\mathbf{x}^-, \mathbf{v}^-) \end{cases}$$

| Implicit Euler

$$\begin{cases} \mathbf{x}^+ = \mathbf{x}^- + h \mathbf{v}^+ \\ \mathbf{M}(\mathbf{v}^+ - \mathbf{v}^-) = h \mathbf{f}(\mathbf{x}^+, \mathbf{v}^+) \end{cases}$$

## | Implicit Euler

$$\begin{cases} \mathbf{x}^+ = \mathbf{x}^- + h \mathbf{v}^+ \\ \mathbf{M}(\mathbf{v}^+ - \mathbf{v}^-) = h \mathbf{f}(\mathbf{x}^+, \mathbf{v}^+) \end{cases}$$



$$\mathbf{r}(\mathbf{x}^+) = \frac{1}{h^2} \mathbf{M}(\mathbf{x}^+ - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x}^+) = \mathbf{0}$$



**Non-Linear  
System of  
Equations**

$$\frac{1}{h^2} \begin{bmatrix} m_1 & & & \\ & m_2 & & \\ & & \ddots & \\ & & & m_n \end{bmatrix} \begin{bmatrix} \mathbf{x}_1^+ - \tilde{\mathbf{x}}_1 \\ \mathbf{x}_2^+ - \tilde{\mathbf{x}}_2 \\ \vdots \\ \mathbf{x}_n^+ - \tilde{\mathbf{x}}_n \end{bmatrix} - \begin{bmatrix} \mathbf{f}_1(\mathbf{x}^+) \\ \mathbf{f}_2(\mathbf{x}^+) \\ \vdots \\ \mathbf{f}_n(\mathbf{x}^+) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

# | Implicit Euler

$$\mathbf{r}(\mathbf{x}^+) = \frac{1}{h^2} \mathbf{M} (\mathbf{x}^+ - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x}^+) = \mathbf{0}$$

Unknown

$\mathbf{x}^- + h \mathbf{v}^-$

Force: Non-Linear Function

Timestep Length

Diagonal Mass Matrix:  $\begin{bmatrix} m_1 & & \\ & \ddots & \\ & & m_n \end{bmatrix}$

The diagram illustrates the Implicit Euler method equation:  $\mathbf{r}(\mathbf{x}^+) = \frac{1}{h^2} \mathbf{M} (\mathbf{x}^+ - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x}^+) = \mathbf{0}$ . Annotations include: 'Unknown' pointing to  $\mathbf{x}^+$ ;  $\mathbf{x}^- + h \mathbf{v}^-$  pointing to  $\tilde{\mathbf{x}}$ ; 'Force: Non-Linear Function' pointing to  $\mathbf{f}(\mathbf{x}^+)$ ; 'Timestep Length' pointing to  $h$ ; and 'Diagonal Mass Matrix' pointing to  $\mathbf{M}$ , which is represented as a diagonal matrix with elements  $m_1, \dots, m_n$ .

# | Newton-Raphson Method

We have:

$$\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

We want:

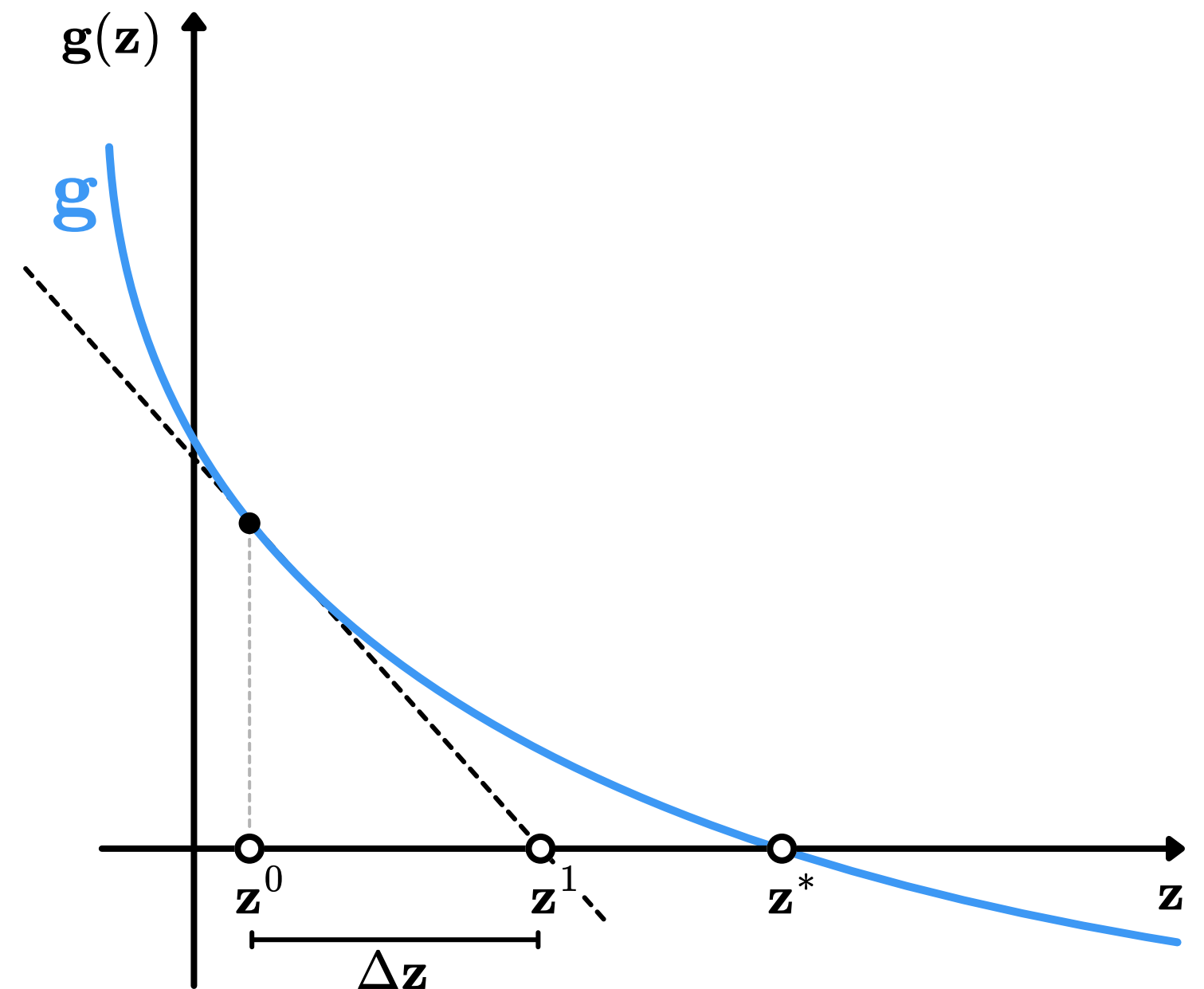
$$\mathbf{g}(\mathbf{z}^*) = \mathbf{0}$$

Linearize around a guess:

$$\mathbf{g}(\mathbf{z}^{(k)} + \Delta \mathbf{z}) \approx \mathbf{g}(\mathbf{z}^{(k)}) + \frac{\partial \mathbf{g}}{\partial \mathbf{z}}(\mathbf{z}^{(k)}) \Delta \mathbf{z} = \mathbf{0}$$

Linear System of Equations

$$\frac{\partial \mathbf{g}}{\partial \mathbf{z}} \Delta \mathbf{z} = -\mathbf{g}(\mathbf{z}^{(k)})$$
$$\mathbf{z}^{(k+1)} = \mathbf{z}^{(k)} + \Delta \mathbf{z}$$



# | Newton-Raphson Method for Implicit Dynamic System

We have:

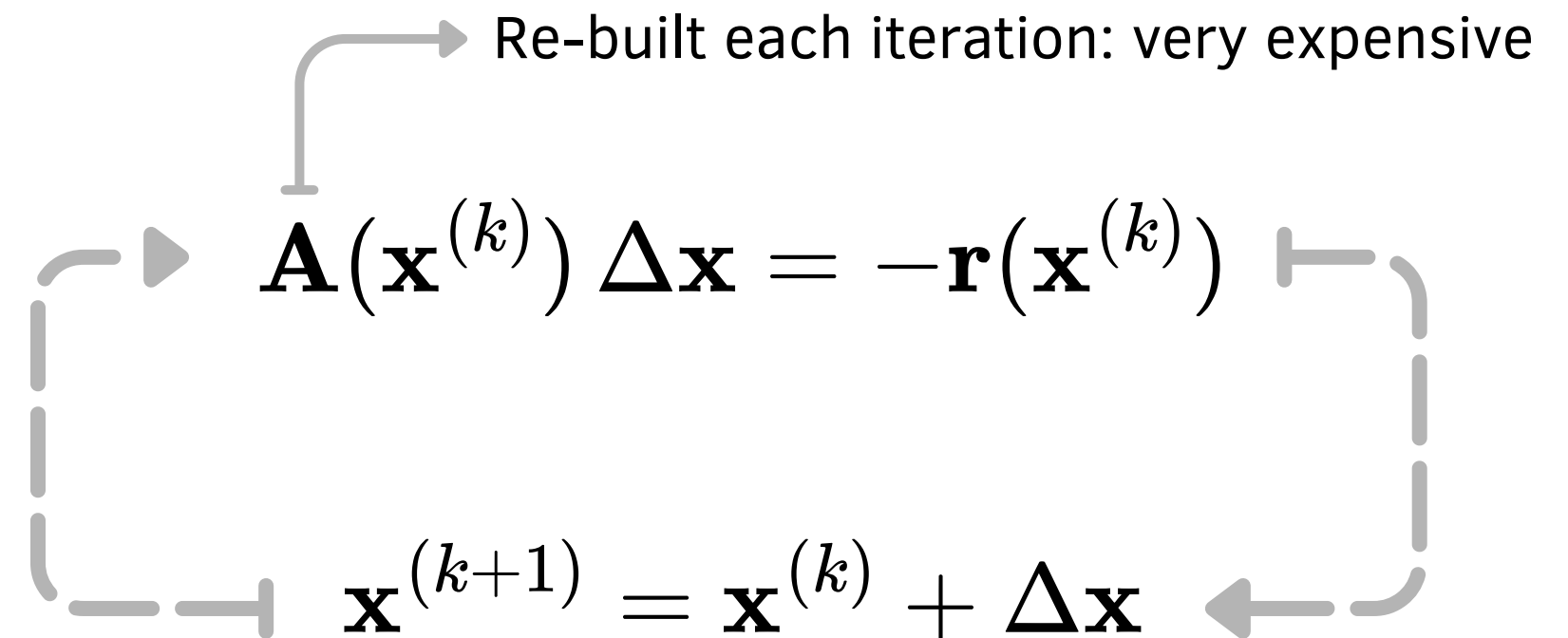
$$\mathbf{r} : \mathbb{R}^{3n} \rightarrow \mathbb{R}^{3n} \quad \text{such that:} \quad \mathbf{r}(\mathbf{x}) = \frac{1}{h^2} \mathbf{M}(\mathbf{x} - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x})$$

We want:

$$\mathbf{r}(\mathbf{x}^+) = \mathbf{0}$$

Linearize around a guess:

$$\frac{\partial \mathbf{r}}{\partial \mathbf{x}} = \underbrace{\frac{1}{h^2} \mathbf{M} - \frac{\partial \mathbf{f}}{\partial \mathbf{x}}}_{\mathbf{A} \in \mathbb{R}^{3n \times 3n}}$$



# | Newton-Raphson Method Single Iteration

Expanded:

$$\underbrace{\left[ \mathbf{M} - h^2 \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right]}_{\mathbf{A}(\mathbf{x}^-)} \Delta \mathbf{x} = \underbrace{h^2 \mathbf{f}(\mathbf{x}^-) + h \mathbf{M} \mathbf{v}^-}_{\mathbf{b}(\mathbf{x}^-)}$$

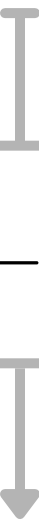
Algorithm:

- 1 **build:**  $\mathbf{A}, \mathbf{b}$
- 2 **solve:**  $\mathbf{A} \Delta \mathbf{x} = \mathbf{b}$
- 3  $\mathbf{x}^+ = \mathbf{x}^- + \Delta \mathbf{x}$
- 4  $\mathbf{v}^+ = \frac{1}{h} (\mathbf{x}^+ - \mathbf{x}^-)$

# | Recasting Root-Finding as Minimization

Does exist a:  $g(\mathbf{x})$  such that:  $\nabla g(\mathbf{x}) = \mathbf{r}(\mathbf{x})$

$$g(\mathbf{x}) = \frac{1}{2h^2} (\mathbf{x} - \tilde{\mathbf{x}})^\top \mathbf{M}(\mathbf{x} - \tilde{\mathbf{x}}) + U(\mathbf{x})$$

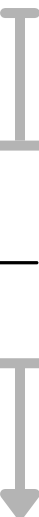

$$\mathbf{f}(\mathbf{x}) = -\frac{\partial U}{\partial \mathbf{x}}(\mathbf{x}) \quad \text{Conservative Forces}$$

$$\nabla g(\mathbf{x}) = \frac{1}{h^2} \mathbf{M}(\mathbf{x} - \tilde{\mathbf{x}}) - \mathbf{f}(\mathbf{x}) = \mathbf{r}(\mathbf{x})$$

# | Recasting Root-Finding as Minimization

Does exists a:  $g(\mathbf{x})$  such that:  $\nabla g(\mathbf{x}) = \mathbf{r}(\mathbf{x})$

$$g(\mathbf{x}) = \frac{1}{2h^2} (\mathbf{x} - \tilde{\mathbf{x}})^\top \mathbf{M} (\mathbf{x} - \tilde{\mathbf{x}}) + U(\mathbf{x})$$


$$\mathbf{f}(\mathbf{x}) = -\frac{\partial U}{\partial \mathbf{x}}(\mathbf{x})$$

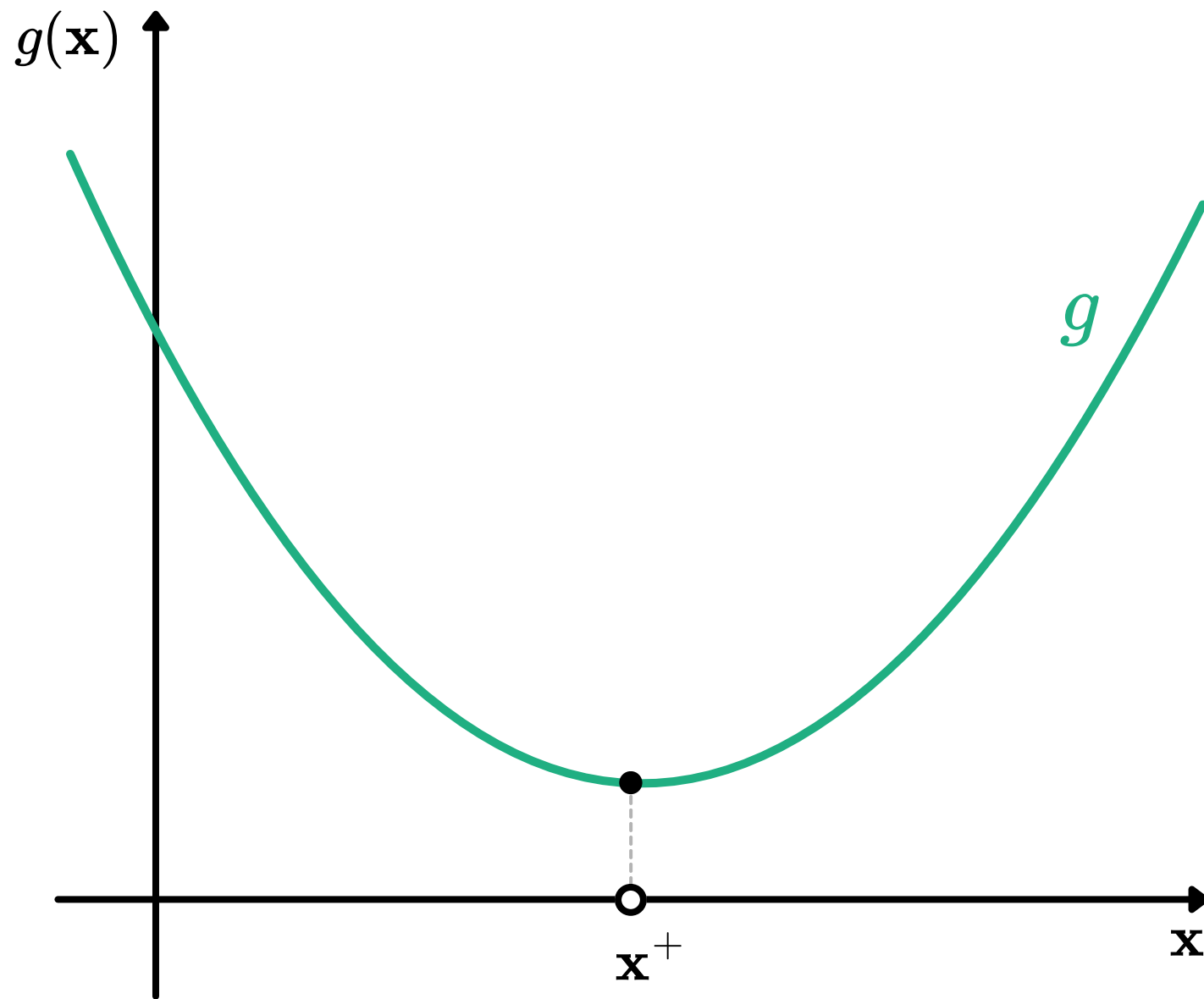
Conservative Forces

$$\mathbf{r}(\mathbf{x}) = \mathbf{0} \quad \Longleftrightarrow \quad \nabla g(\mathbf{x}) = \mathbf{0} \quad \begin{array}{c} \text{if Convex} \\ \Longleftrightarrow \end{array} \quad \min_{\mathbf{x}} g(\mathbf{x})$$

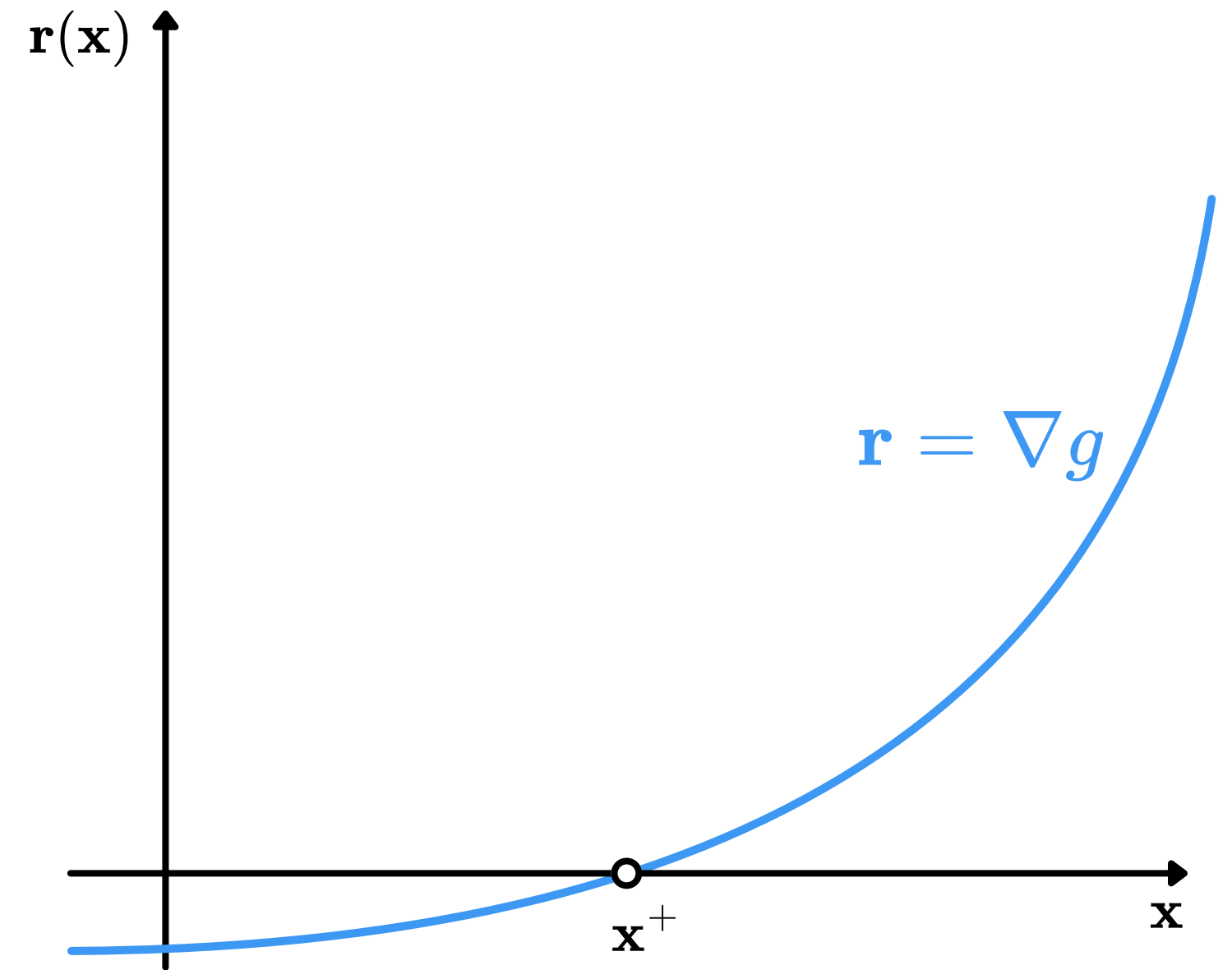
# | Root-Finding and Minimization Equivalence

Equivalent Problems:

$$\mathbf{x}^+ = \arg \min_{\mathbf{x}} g(\mathbf{x})$$



$$\mathbf{r}(\mathbf{x}^+) = 0$$



# | Descent Methods How to Solve the Minimization Problem

```
1 Initialize  $\mathbf{x}^{(0)}$ 
2 for  $k = 0, 1, \dots, N - 1$  do
3    $\Delta \mathbf{x} = \text{DescentDirection}()$ 
4    $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \cdot \Delta \mathbf{x}$ 
5  $\mathbf{x}^+ = \mathbf{x}^{(N)}$ 
6  $\mathbf{v}^+ = \frac{1}{h}(\mathbf{x}^+ - \mathbf{x}^-)$ 
```

| Jacobian:

$$\mathbf{J} = \nabla g(\mathbf{x}) = \mathbf{r}(\mathbf{x}) \in \mathbb{R}^{3n}$$

| Hessian:

$$\mathbf{H} = \nabla^2 g(\mathbf{x}) = \mathbf{A}(\mathbf{x}) \in \mathbb{R}^{3n \times 3n}$$

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| Gradient Descent:

$$\Delta \mathbf{x}^{(k)} = -\mathbf{J}^{(k)}$$

| Newton:

$$\Delta \mathbf{x}^{(k)} = -[\mathbf{H}^{(k)}]^{-1} \mathbf{J}^{(k)}$$

| Quasi-Newton:

$$\Delta \mathbf{x}^{(k)} = -\mathbf{P}^{(k)} \mathbf{J}^{(k)}, \quad \mathbf{P}^{(k)} \approx [\mathbf{H}^{(k)}]^{-1}$$

| Projective Dynamics:

$$\Delta \mathbf{x}^{(k)} = -\mathbf{L}_{\text{PD}}^{-1} \mathbf{J}_{\text{PD}}^{(k)}$$

$$\mathbf{J}_{\text{PD}} = \nabla g_{\text{PD}}(\mathbf{x}), \quad \mathbf{L}_{\text{PD}} \approx \mathbf{H} \text{ (fixed)}$$

| Baraff-Witkin:

$$\Delta \mathbf{x} = -\mathbf{H}^{-1} \mathbf{J}, \quad k = 1$$

# | Newton's Minimization Method = Newton-Raphson Root Finding's Method

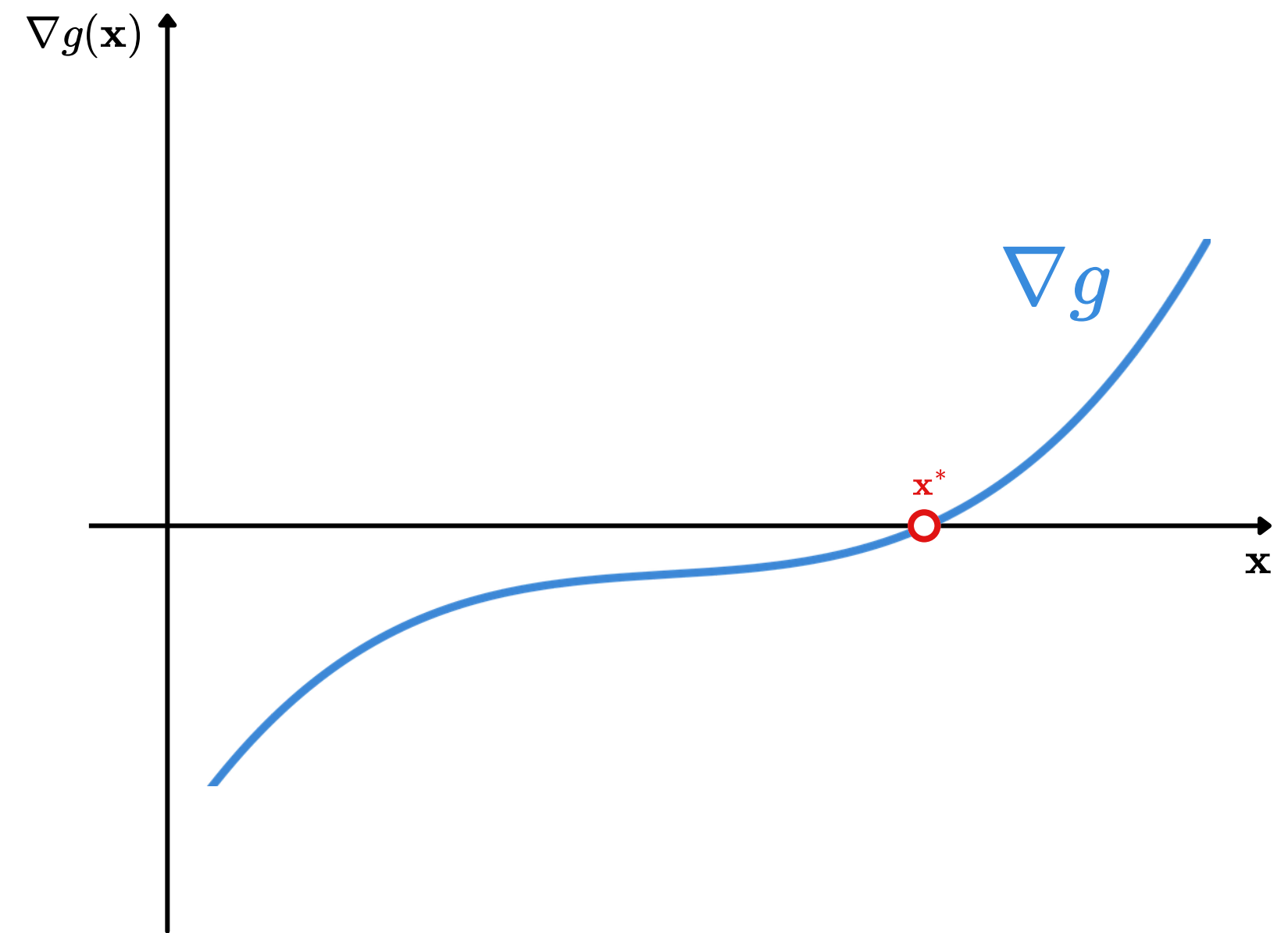
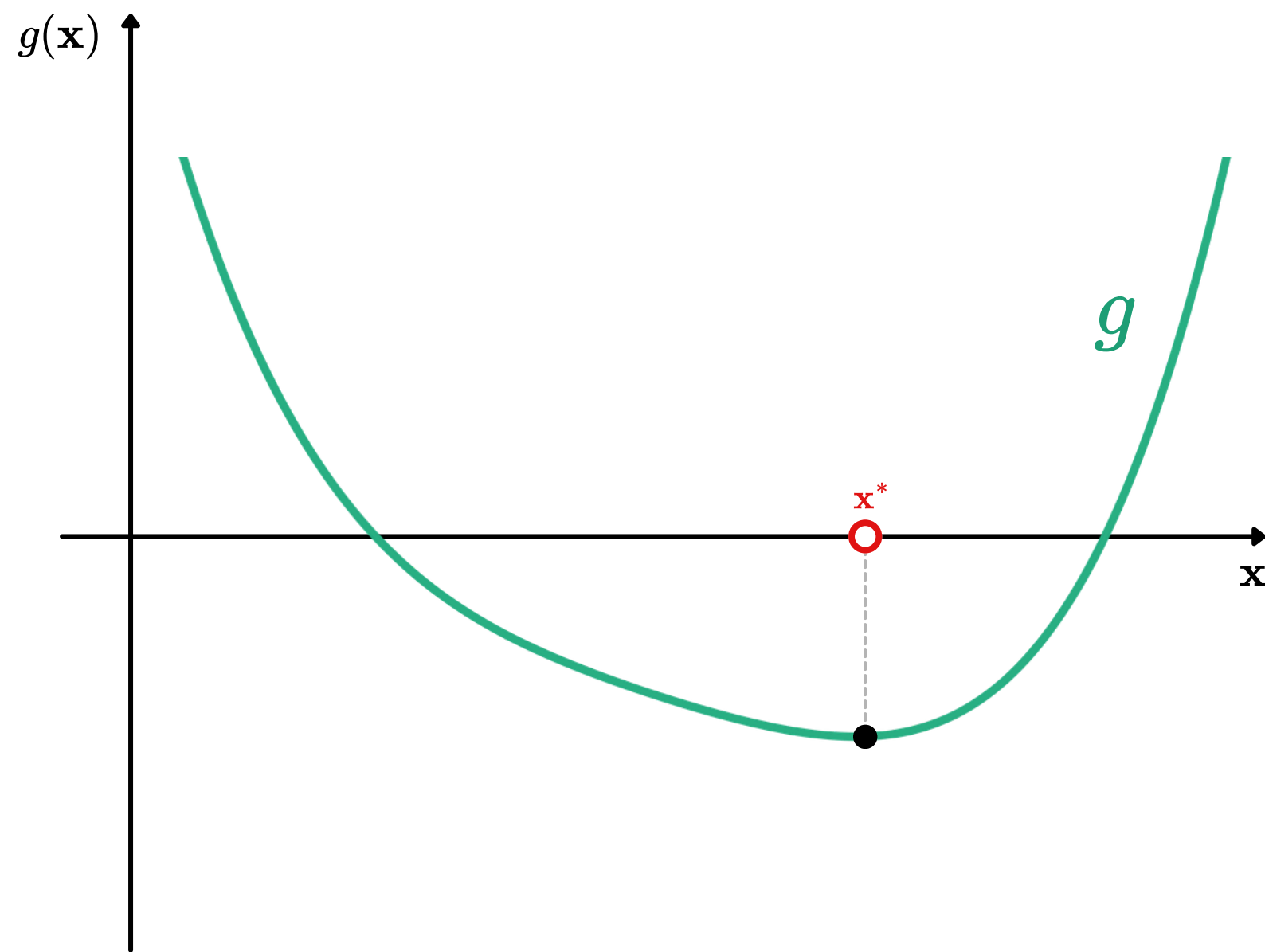
**function** NEWTONMETHOD( $\mathbf{x}^-$ ,  $\mathbf{v}^-$ ,  $N$ )

```
1 Initialize  $\mathbf{x}^{(0)}$ 
2 for  $k = 0, 1, \dots, N - 1$  do
3     BUILD  $\mathbf{J}^{(k)}$ ,  $\mathbf{H}^{(k)}$ 
4     SOLVE  $\mathbf{H}^{(k)} \Delta \mathbf{x} = -\mathbf{J}^{(k)}$ 
5      $\mathbf{x}^{(k+1)} \leftarrow \mathbf{x}^{(k)} + \Delta \mathbf{x}$ 
6  $\mathbf{x}^+ \leftarrow \mathbf{x}^{(N)}$ 
7  $\mathbf{v}^+ \leftarrow \frac{1}{h}(\mathbf{x}^+ - \mathbf{x}^-)$ 
```

**function** NEWTONRAPHSON( $\mathbf{x}^-$ ,  $\mathbf{v}^-$ ,  $N$ )

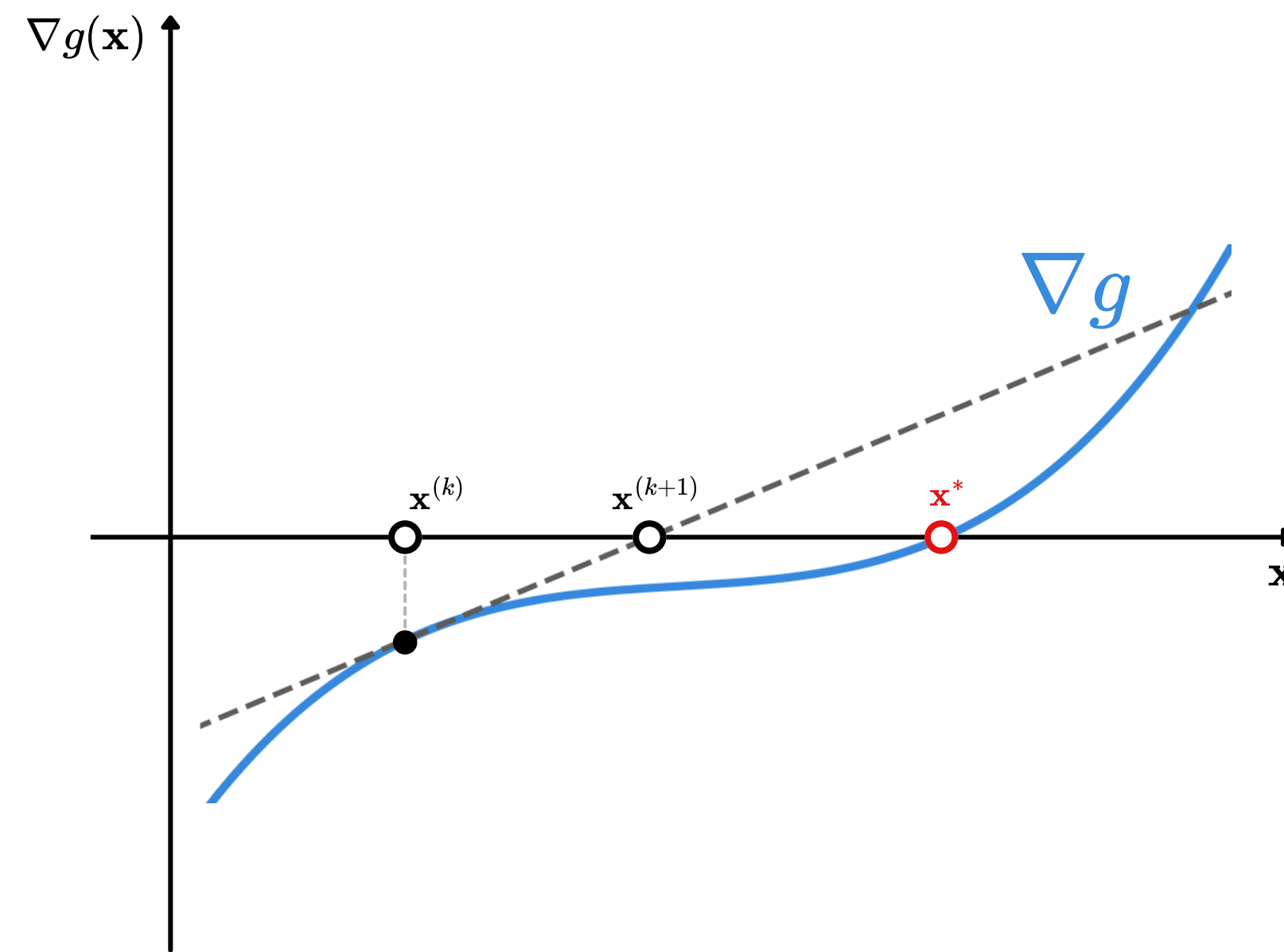
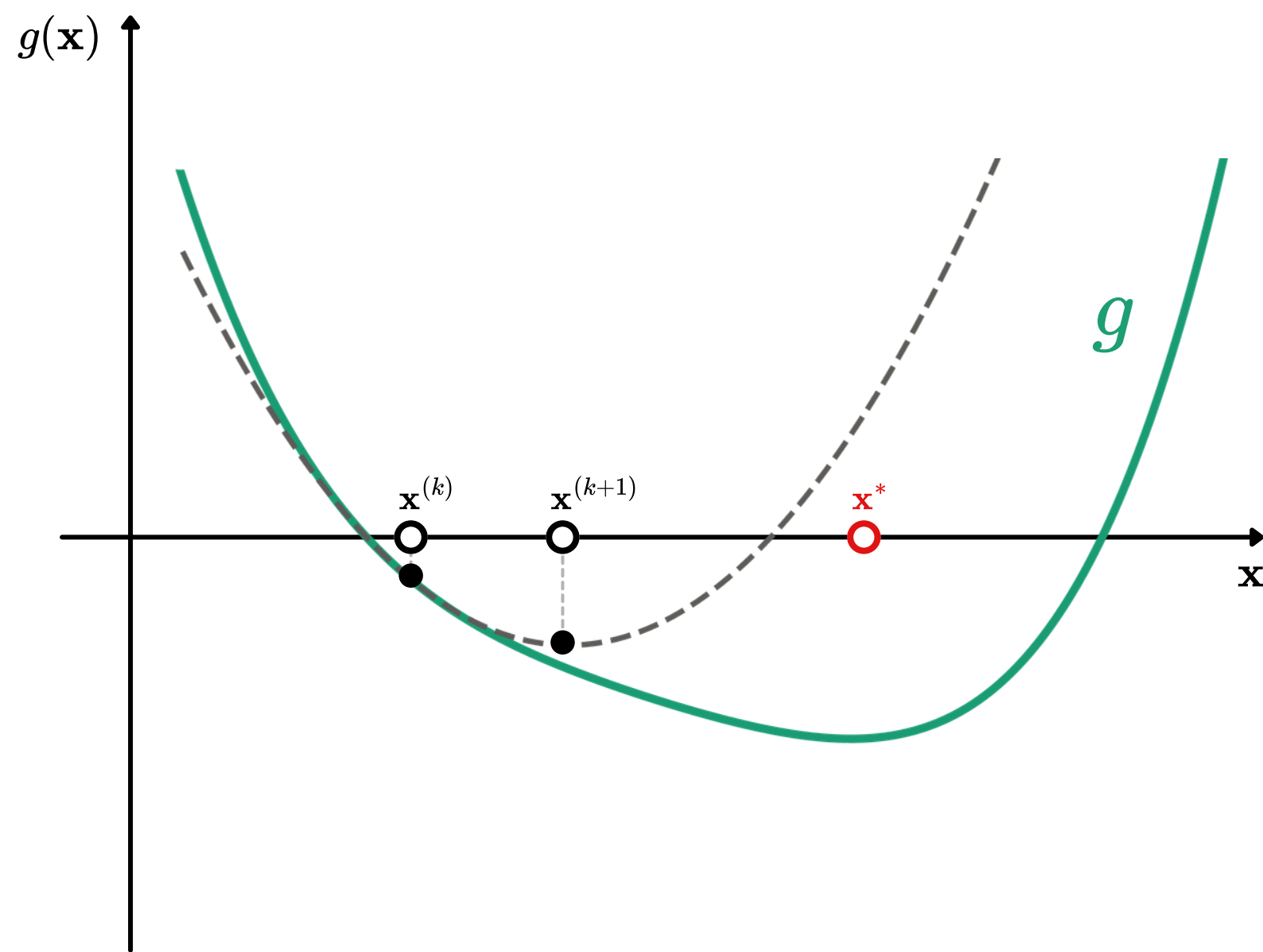
```
1 Initialize  $\mathbf{x}^{(0)}$ 
2 for  $k = 0, 1, \dots, N - 1$  do
3     BUILD  $\mathbf{A}^{(k)}$ ,  $\mathbf{b}^{(k)}$ 
4     SOLVE  $\mathbf{A}^{(k)} \Delta \mathbf{x} = \mathbf{b}^{(k)}$ 
5      $\mathbf{x}^{(k+1)} \leftarrow \mathbf{x}^{(k)} + \Delta \mathbf{x}$ 
6  $\mathbf{x}^+ \leftarrow \mathbf{x}^{(N)}$ 
7  $\mathbf{v}^+ \leftarrow \frac{1}{h}(\mathbf{x}^+ - \mathbf{x}^-)$ 
```

# | Convexity and Newton's Method When the Energy **IS** Convex



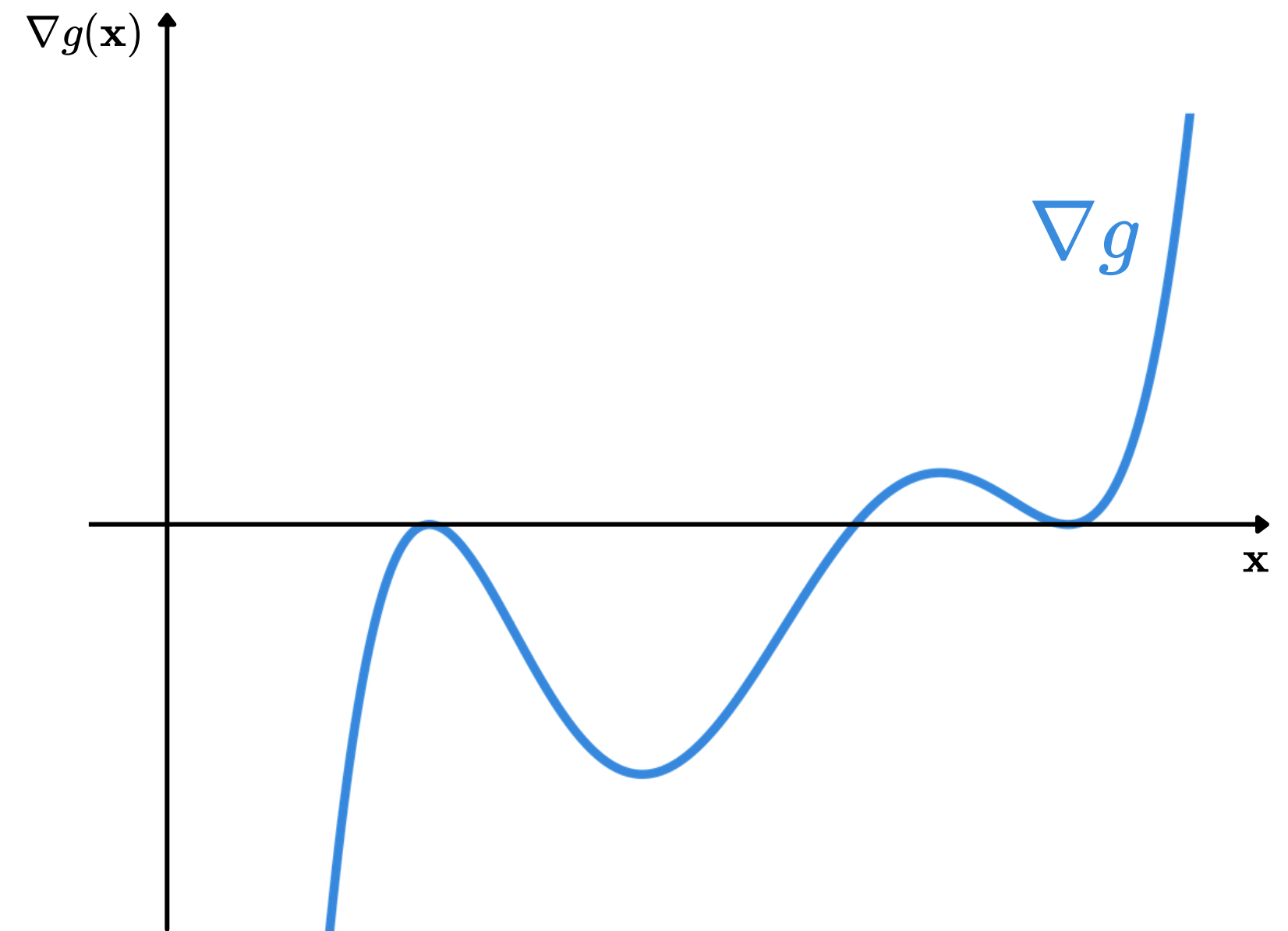
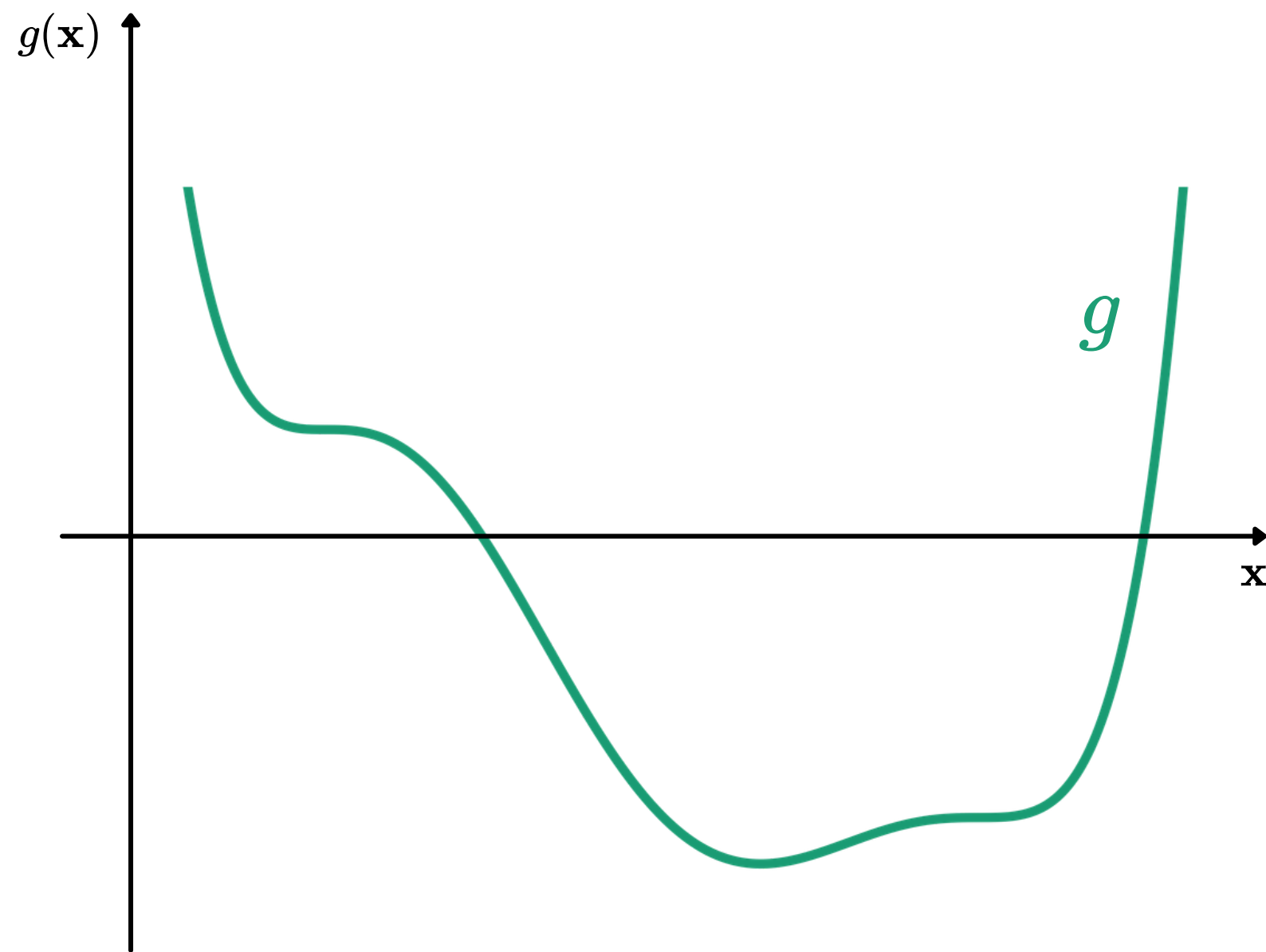
# | Convexity and Newton's Method When the Energy **IS** Convex

$$g(\mathbf{x}^{(k+1)}) < g(\mathbf{x}^{(k)})$$

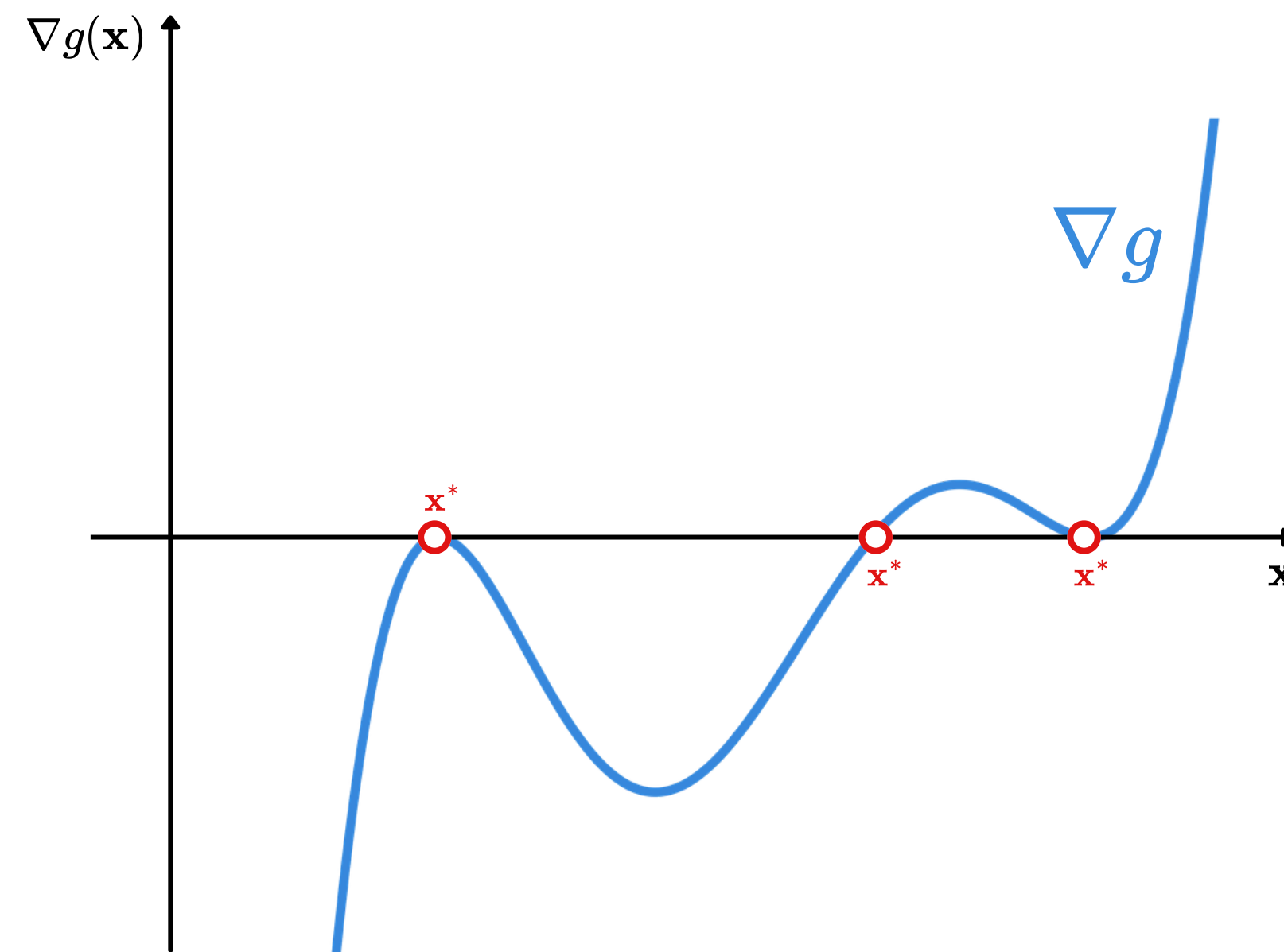
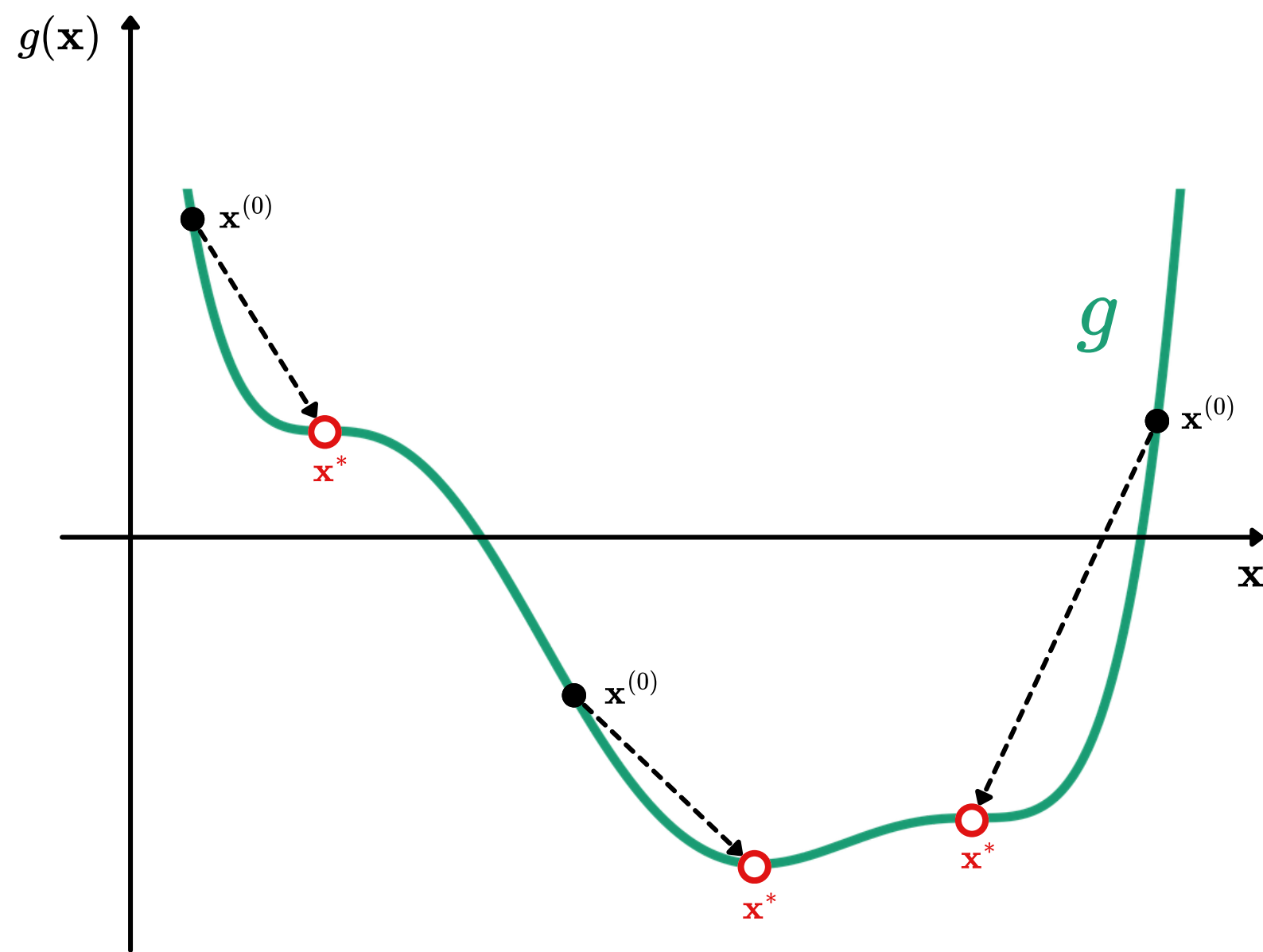


# | Convexity and Newton's Method When the Energy **IS NOT** Convex

- There might be multiple stationary points
- There might be areas where the curvature is negative



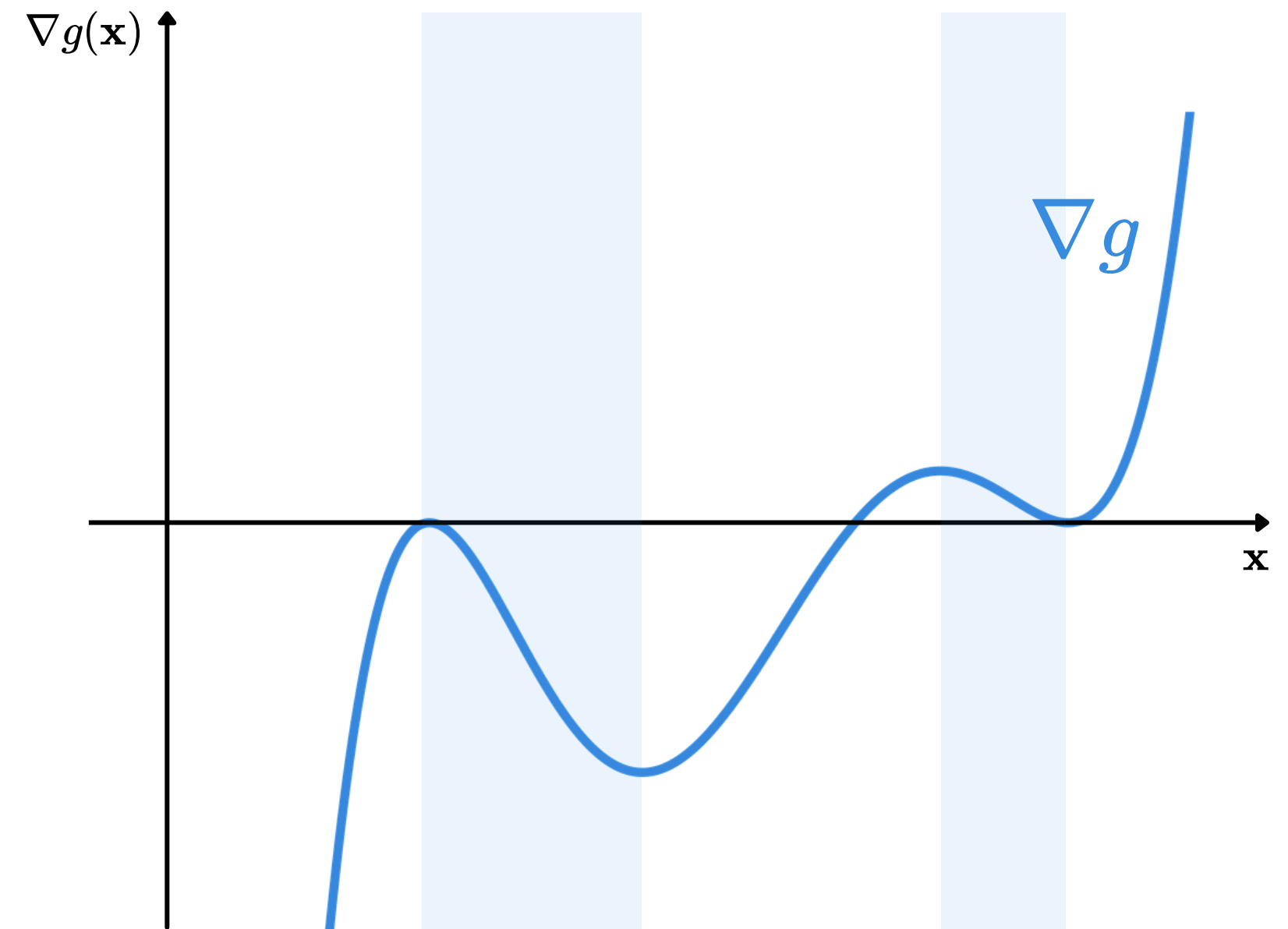
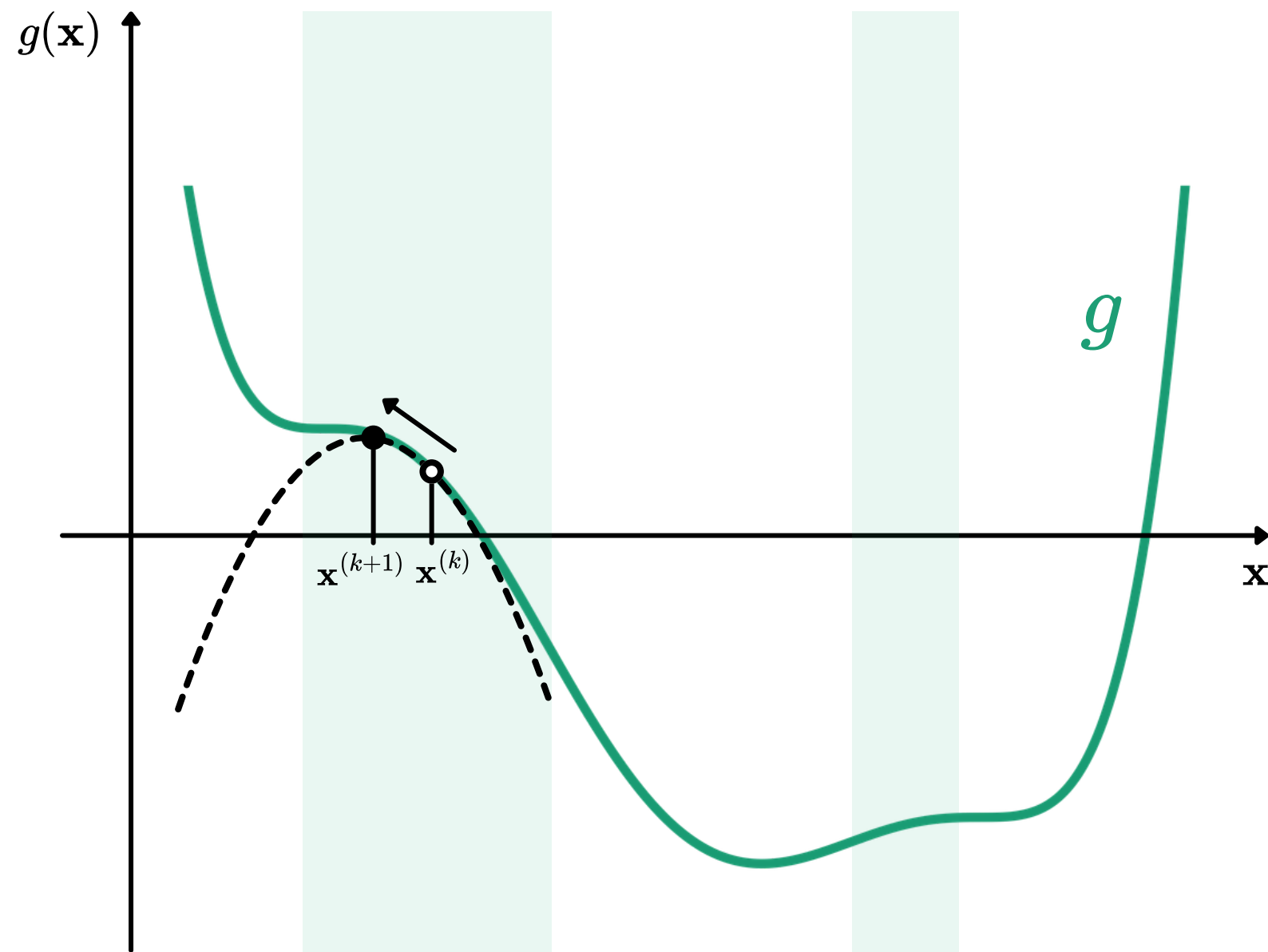
# | Convexity and Newton's Method: Multiple Stationary Points



# | Convexity and Newton's Method: Negative Curvature

If the curvature is negative, by taking the Newton's step, there might be an **increase** in Energy

$$g(\mathbf{x}^{(k+1)}) > g(\mathbf{x}^{(k)})$$



# | Convexity and Newton's Method: Solve the Negative Curvature Problem

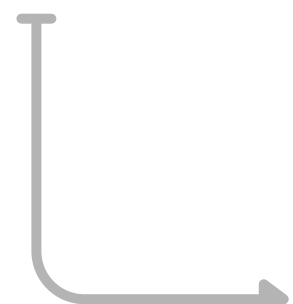
A matrix is SPD (**S**ymmetric **P**ositive **D**efinite), written  $\mathbf{A} \succ 0$  if it is symmetric and all its eigenvalues are positive:

$$\mathbf{A} = \mathbf{A}^\top \quad \text{and} \quad \mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top, \quad \lambda_i > 0 \quad \forall i.$$

- $\mathbf{A} \succ 0, \mathbf{B} \succ 0 \Rightarrow \mathbf{A} + \mathbf{B} \succ 0$
- $\mathbf{A} \succ 0, \mathbf{B} \not\succ 0 \Rightarrow \mathbf{A} + \mathbf{B}$  no guarantee

$\mathbf{H} \not\succ 0$  :  $\Delta \mathbf{x} = -\mathbf{H}^{-1} \mathbf{J}$  then  $\Delta \mathbf{x}$  may not be a descent direction

$\bar{\mathbf{H}} \succ 0$  :  $\Delta \mathbf{x} = -\bar{\mathbf{H}}^{-1} \mathbf{J}$  then  $\alpha \cdot \Delta \mathbf{x}$  is guaranteed to be a descent direction for a sufficiently small  $\alpha > 0$

 is the SPD-Projection of  $\mathbf{H}$

# Convexity and Newton's Method: Solve the Negative Curvature Problem

How to fix the Energy's Hessian

Always Convex

$$g(\mathbf{x}) = \frac{1}{2h^2} (\mathbf{x} - \tilde{\mathbf{x}})^\top \mathbf{M} (\mathbf{x} - \tilde{\mathbf{x}}) + U(\mathbf{x})$$

$\mathbf{H} = \frac{1}{h^2} \mathbf{M} + \nabla^2 U(\mathbf{x})$

Always SPD      Might not be SPD:  
we must **SPD-Project**

$\nabla^2 U(\mathbf{x}) \in \mathbb{R}^{3n \times 3n}$  : it is very expensive to SPD-Project such a big matrix

# Convexity and Newton's Method: Solve the Negative Curvature Problem

How is the Potential Energy Factor Constructed

$$U(\mathbf{x}) = \sum_e U_e(\mathbf{x}_e)$$

The total energy is the sum of single element energies. Each element (edge, triangle, tetrahedron) involves only a few particles (2, 3 or 4).

$$\nabla^2 U(\mathbf{x}) = \sum_e \nabla^2 U_e(\mathbf{x}_e)$$

$$\nabla^2 U_e(\mathbf{x}_e) \in \mathbb{R}^{6 \times 6 | 9 \times 9 | 12 \times 12}$$

So the Hessian is also a sum of small element Hessians. Each one can be easily SPD-projected independently and in parallel, then assembled into the full Hessian.

# Convexity and Newton's Method: Solve the Negative Curvature Problem

How an SPD-Projection is computed

How do we SPD-Project the single element energies:

$$\nabla^2 U_e(\mathbf{x}_e) \in \mathbb{R}^{6 \times 6 | 9 \times 9 | 12 \times 12}$$

Two ways:

- **Numerical:**

$$\nabla^2 U_e = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top \longrightarrow \bar{\lambda}_i = \max(\lambda_i, 0) \longrightarrow \overline{\nabla^2 U_e} = \mathbf{Q} \bar{\mathbf{\Lambda}} \mathbf{Q}^\top$$

- **Analytical:**

for many frequently used energies the eigenvalues and eigenvectors are known in closed form. This route is usually faster and more robust.

# | Analytical SPD Projection Quadratic Spring Potential

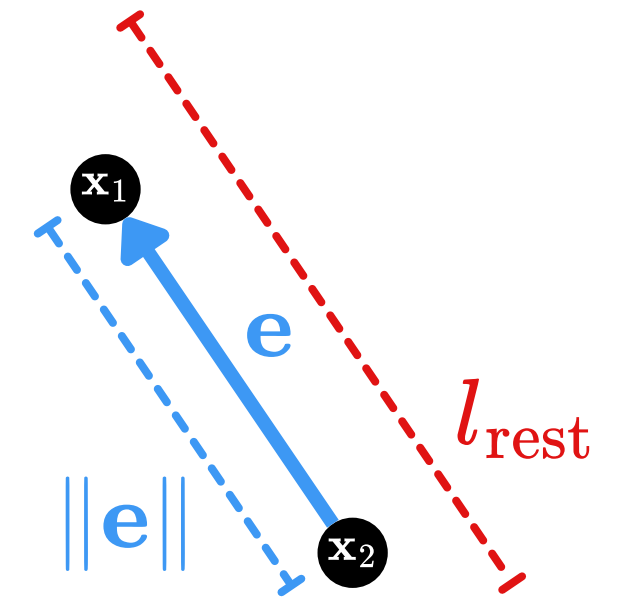
**Spring:**

$$\mathbf{x}_e = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} \in \mathbb{R}^6 \quad \mathbf{e} = \mathbf{x}_1 - \mathbf{x}_2$$

$$\hat{\mathbf{e}} = \frac{\mathbf{e}}{\|\mathbf{e}\|}$$

**Energy:**

$$U_e(\mathbf{x}_e) = \frac{k}{2} (\|\mathbf{e}\| - l_{\text{rest}})^2$$



**Hessian:**

$$\mathbf{H}_e = \begin{bmatrix} \mathbf{K} & -\mathbf{K} \\ -\mathbf{K} & \mathbf{K} \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

$$\mathbf{K} = k \left[ \hat{\mathbf{e}} \hat{\mathbf{e}}^\top + \left( 1 - \frac{l}{\|\mathbf{e}\|} \right) (\mathbf{I} - \hat{\mathbf{e}} \hat{\mathbf{e}}^\top) \right]$$

**Eigenvalues:**

$$\lambda_1 \quad \lambda_2 \quad \lambda_3 \quad \lambda_4 \quad \lambda_5 = \lambda_6$$

$$0, \quad 0, \quad 0, \quad 2k, \quad 2k \left( 1 - \frac{l}{\|\mathbf{e}\|} \right)$$

Can be negative

**SPD-Hessian:**

$$\bar{\mathbf{H}}_e = \begin{bmatrix} \bar{\mathbf{K}} & -\bar{\mathbf{K}} \\ -\bar{\mathbf{K}} & \bar{\mathbf{K}} \end{bmatrix} \in \mathbb{R}^{6 \times 6} \quad \bar{\mathbf{H}}_e \succeq 0$$

$$\bar{\mathbf{K}} = k \left[ \hat{\mathbf{e}} \hat{\mathbf{e}}^\top + \max \left( 1 - \frac{l}{\|\mathbf{e}\|}, 0 \right) (\mathbf{I} - \hat{\mathbf{e}} \hat{\mathbf{e}}^\top) \right]$$

# | SPD-Aware Newton's Method Algorithm

**function** SPDAWARENEWTONMETHOD( $\mathbf{x}^-$ ,  $\mathbf{v}^-$ )

2: Initialize  $\mathbf{x}^{(0)}$

3: **for**  $k = 0, 1, \dots, N - 1$  **do**

4:      $\mathbf{J} \leftarrow \nabla g(\mathbf{x}^{(k)})$

5:      $\bar{\mathbf{H}} \leftarrow \text{SPDPROJECTION}(\nabla^2 g(\mathbf{x}^{(k)}))$

6:     SOLVE  $\bar{\mathbf{H}} \Delta \mathbf{x} = -\mathbf{J}$

7:      $\alpha \leftarrow \text{LINESEARCH}(\mathbf{x}^{(k)}, \Delta \mathbf{x}, \mathbf{J})$

8:      $\mathbf{x}^{(k+1)} \leftarrow \mathbf{x}^{(k)} + \alpha \Delta \mathbf{x}$

9:  $\mathbf{x}^+ \leftarrow \mathbf{x}^{(N)}$

10:  $\mathbf{v}^+ \leftarrow \frac{1}{h}(\mathbf{x}^+ - \mathbf{x}^-)$

**end function**